

Radicals

In this session¹¹ we are going to tackle the limits for radical functions. Recalling that a rational function has the following algebraic expression:

$$f(x) = a\sqrt{bx+h} + k.$$

Let's remember that the domain of this functions are such that $bx+h \geq 0 \rightarrow x \geq -h/b$. This means that if it is asked to compute a limit that is less than $-h/b$ the limit is not defined. Evemore, the limit to that value only defined when approaching from $\pm\infty$, depending on the sign of b . For example,

$$\lim_{x \rightarrow 1} \sqrt{x-1} = \sqrt{1-1} = \sqrt{0} = 0.$$

This is consistent, since $x \geq 0$. If we evaluate,

$$\lim_{x \rightarrow 0.5} \sqrt{x-1} = \sqrt{0.5-1} = \sqrt{-1/2}.$$

In this case $x < 0$, therefore the limit does not exist.

As a final example, let's see the limits of $\sqrt{x}$ from $\pm\infty$ as $x \rightarrow 0$,

$$\lim_{x \rightarrow 0^-} \sqrt{x} = \text{DNE}$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

Another interesting function to analyse limits is when we have a rational function of radical functions, as follows

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x}.$$

In order to apply the direct substitution method, we need to make algebraic modifications. In this case we are going to use the powerfull technique of multiplying by 1, made from the conjugate, that is,

$$1 = \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}.$$

Let's multiply by 1 the function¹² and we get,

$$\frac{\sqrt{x+4}-2}{x} = \frac{1}{\sqrt{x+4}+2}$$

Now, let's try the direct substitution after this algebraic manipulation,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x} &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4}+2} \\ &= \frac{1}{\sqrt{(0)+4}+2} \\ &= \frac{1}{4} \end{aligned}$$

¹¹ January 28

Class Activity Compute the limits of the following functions by using the analytical methods seen in this session.

$$\begin{aligned} &\lim_{x \rightarrow -1} \sqrt{-x} \\ &\lim_{x \rightarrow 1/2} \sqrt{2x + \frac{1}{2}} \\ &\lim_{x \rightarrow 1} \sqrt{x-2} \\ &\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2} \end{aligned}$$

¹²

$$\begin{aligned} \frac{\sqrt{x+4}-2}{x} &= \frac{\sqrt{x+4}-2}{x} \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2} \\ &= \frac{(x+4)-4}{x\sqrt{x+4}+2} \\ &= \frac{x}{x\sqrt{x+4}+2} \\ &= \frac{1}{\sqrt{x+4}+2} \end{aligned}$$